

AWS Invoice Pipeline — API Reference

Complete REST API specification for the AWS Invoice Processing Pipeline. All endpoints are served via **Amazon API Gateway** and route to the `invoice-ingestion` or `approval-handler` Lambda functions.

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Base Architecture

Property	Value
Protocol	HTTPS / REST
API Name	InvoicePipelineAPI-{env}
Stage	Determined by the Environment SAM parameter (dev · staging · prod)
Base URL	https://<api-id>.execute-api.<region>.amazonaws.com/<env>
Content-Type	application/json (except email callback endpoints which return text/html)
CORS	Enabled for all origins (*), supporting GET, POST, PUT, DELETE, OPTIONS
Throttling	API Gateway default: 10,000 requests/sec (account level), 5,000 burst

Lambda Routing

Lambda Function	Endpoints Handled
invoice-ingestion-{env}	POST /upload
approval-handler-{env}	All other endpoints (/approve, /reject, /approvals, /invoices, /exceptions, /audit, /dashboard)

Authentication & Security

Current implementation: No API key or IAM authentication is enforced on the API Gateway endpoints. Access is controlled at the network level and via CORS headers only.

Aspect	Detail
API Keys	Not required
IAM Auth	Not enabled on API Gateway resources
CORS	Fully open (* origin) — suitable for development. Restrict AllowOrigin to your Amplify domain in production.
SES Email Auth	Approval email links contain a Step Functions task token as a query parameter. The token acts as a one-time credential — it expires after the workflow completes or times out (72 hours).

Common Headers

Request Headers

Header	Required	Description
Content-Type	Yes (for POST)	application/json or application/pdf (for upload)
X-Filename	No	Original filename of the uploaded PDF (used by POST /upload)
X-Invoice-Metadata	No	JSON string with optional pre-filled fields: vendorName, invoiceDate, poNumber, invoiceNumber, uploadedBy

Response Headers (All Endpoints)

All responses include the following CORS headers:

```
Access-Control-Allow-Origin: *
Access-Control-Allow-Headers: Content-Type,Authorization,X-Requested-With,x-
filename,x-invoice-metadata,X-Filename,X-Invoice-Metadata
Access-Control-Allow-Methods: GET,POST,PUT,DELETE,OPTIONS
Content-Type: application/json
```

CORS Preflight

Every API path supports `OPTIONS` for CORS preflight. Preflight requests return `200 OK` with CORS headers and an empty body.

✖ Standard Error Response

All error responses follow a consistent JSON structure:

```
{
  "error": "Human-readable error description",
  "message": "Detailed error message (optional, included for 500 errors)"
}
```

Common HTTP Status Codes

Code	Meaning	When Returned
200	Success	All successful requests
400	Bad Request	Missing required parameters (invoiceId, token, action, etc.)
404	Not Found	Invoice not found, or unmatched API path
405	Method Not Allowed	Non-POST request to /upload
500	Internal Server Error	Unhandled exceptions — includes error and message fields

Invoice Status Lifecycle

```
stateDiagram-v2
    [*] --> IN_PROGRESS : Invoice ingested
    IN_PROGRESS --> PROCESSED : No anomalies (auto-approved)
    IN_PROGRESS --> EXCEPTION : High-severity anomaly detected
    IN_PROGRESS --> IN_REVIEW : Medium-severity anomaly detected

    EXCEPTION --> PENDING : Approval email sent (vendorId = PENDING)
    IN_REVIEW --> PENDING : Approval email sent

    PENDING --> PROCESSED : Reviewer approves
    PENDING --> RESOLVED : Reviewer rejects
    PENDING --> IN_PROGRESS : Reviewer requests reprocess

    PROCESSED --> [*]
    RESOLVED --> [*]
```

Status Reference

Status	Badge Color	Meaning
IN_PROGRESS	 Orange	Currently being processed through the pipeline
PROCESSED	 Green	Successfully validated and persisted — no anomalies
IN_REVIEW	 Orange	Medium-severity anomaly — awaiting human review
PENDING_REVIEW	 Yellow	Queued for review but not yet assigned
EXCEPTION	 Red	High-severity anomaly detected — requires immediate attention

RESOLVED	 Blue	Exception reviewed and resolved (approved or rejected)
----------	--	--

Endpoints

1. Ingestion — POST /upload

Directly uploads an invoice PDF into the pipeline via the React dashboard.

Lambda: `invoice-ingestion-{env}`

Request

```
curl -X POST "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/upload" \
-H "Content-Type: application/pdf" \
-H "X-Filename: invoice-acme-2026.pdf" \
-H "X-Invoice-Metadata: {\"vendorName\":\"Acme Corp\",\"poNumber\":\"PO-1234\"}" \
--data-binary @invoice.pdf
```

Header	Required	Description
Content-Type	Yes	application/pdf (binary body is base64-encoded by API Gateway)
X-Filename	No	Original filename. Defaults to <code>invoice.pdf</code> if omitted.
X-Invoice-Metadata	No	JSON string with optional pre-filled metadata fields (see below)

X-Invoice-Metadata fields (all optional):

Field	Type	Description
vendorName	string	Pre-filled vendor name
vendorId	string	Pre-filled vendor ID
invoiceNumber	string	Pre-filled invoice number
invoiceDate	string	Pre-filled invoice date
poNumber	string	Purchase order number
uploadedBy	string	Name of the user performing the upload

Success Response — 200 OK

```
{
  "success": true,
  "invoiceId": "INV-2026-A1B2C3D4",
  "message": "Invoice uploaded and processing started",
  "s3Key": "invoices/INV-2026-A1B2C3D4/invoice-acme-2026.pdf"
}
```

Field	Type	Description
success	boolean	Always true on success
invoiceId	string	Generated ID in format INV-{YEAR}-{8-CHAR-UUID}
message	string	Human-readable confirmation
s3Key	string	S3 object key where the PDF was stored

Error Responses

Code	Body	Cause
405	{ "error": "Method not allowed" }	Non-POST request
500	{ "error": "Upload failed", "message": "... " }	S3 upload failure, DynamoDB write failure, or Step Functions start failure

Side Effects

1. PDF stored in S3 bucket `invoice-pipeline-raw-{accountId}` at key `invoices/{invoiceId}/{filename}`
2. InvoiceRecord created in DynamoDB with status `IN_PROGRESS` and source `UPLOAD`
3. AuditEntry written with eventType `INGESTION`
4. Step Functions execution started with the `invoiceId`, `s3Bucket`, and `s3Key`

2. Email Callbacks — GET /approve · GET /reject

Triggered via hyperlinks in SES approval emails. These endpoints resume the paused Step Functions workflow.

Lambda: `approval-handler-{env}`

GET /approve

```
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/approve?token=<taskToken>&invoiceId=INV-2026-A1B2C3D4"
```

GET /reject

```
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/reject?token=<taskToken>&invoiceId=INV-2026-A1B2C3D4"
```

Query Parameters

Parameter	Required	Description
token	Yes	Step Functions task token (URL-encoded). Acts as a one-time credential.
invoiceId	Yes	ID of the invoice being reviewed

Success Response — 200 OK

Returns a rendered **HTML page** (not JSON) confirming the action. The `Content-Type` header is set to `text/html`.

The HTML page displays:

-  or  icon depending on action
- Confirmation message: "Invoice **{invoiceId}** has been approved/rejected successfully."
- "You can close this window." footer text

Error Responses

Code	Body	Cause
400	{ "error": "Missing token or invoiceId" }	Missing required query parameters
500	{ "error": "Failed to process callback" }	Step Functions SendTaskSuccess/SendTaskFailure call failed (e.g., token expired)

Side Effects

Action	DynamoDB Update	Step Functions	Audit Entry
Approve	Status → PROCESSED, sets approvedBy: "email-callback"	SendTaskSuccess with approval payload	APPROVAL event
Reject	Status → RESOLVED, sets approvalComments: "Rejected via email link"	SendTaskFailure with error REJECTED	REJECTION event

3. Dashboard — GET /dashboard

Retrieves aggregated statistics across the entire invoice database to power dashboard KPI cards.

Lambda: `approval-handler-{env}`

Request

```
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/dashboard"
```

No query parameters required.

Success Response — 200 OK

```
{
  "totalInvoices": 150,
  "processed": 120,
  "inProgress": 10,
  "exceptions": 15,
  "inReview": 3,
  "pendingReview": 2,
  "resolved": 0,
  "processedPercentage": "80.0",
```

```
"exceptionPercentage": "10.0"
}
```

Field	Type	Description
totalInvoices	number	Sum of all invoices across all statuses
processed	number	Count of PROCESSED invoices
inProgress	number	Count of IN_PROGRESS invoices
exceptions	number	Count of EXCEPTION invoices
inReview	number	Count of IN_REVIEW invoices
pendingReview	number	Count of PENDING_REVIEW invoices
resolved	number	Count of RESOLVED invoices
processedPercentage	string	(processed / total × 100) with 1 decimal place
exceptionPercentage	string	(exceptions / total × 100) with 1 decimal place

Implementation note: Each status count is fetched via a separate DynamoDB query on the `status-index` GSI. All 6 queries run in parallel for performance.

4. Invoices — GET /invoices · GET /invoices/{invoiceId}

GET /invoices

Retrieves a paginated list of all invoices with optional status filtering.

```
# All invoices (default page size 25)
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/invoices"

# Filtered by status with custom page size
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/invoices?
status=PROCESSED&pageSize=10"

# Pagination (pass lastKey from previous response)
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/invoices?
lastKey=%7B%22invoiceId%22%3A%22INV-2026-A1B2C3D4%22%7D"
```

Query Parameters

Parameter	Required	Default	Description
status	No	—	Filter by InvoiceStatus: IN_PROGRESS, PROCESSED, IN_REVIEW, PENDING_REVIEW, EXCEPTION, RESOLVED
pageSize	No	25	Maximum number of results per page
lastKey	No	—	URL-encoded JSON string of DynamoDB LastEvaluatedKey for cursor-based pagination

Success Response — 200 OK

```
{
  "items": [
    {
      "invoiceId": "INV-2026-A1B2C3D4",
      "vendorId": "PENDING",
      "vendorName": "Acme Solutions Pvt. Ltd.",
      "vendorAddress": "123 MG Road, Bengaluru 560001",
      "gstIn": "29AABCU9603R1ZM",
      "invoiceNumber": "INV-2026-9812",
      "invoiceDate": "2026-06-15",
      "dueDate": "2026-07-15",
      "poNumber": "PO-4567",
      "lineItems": [
        {
          "description": "AWS Training - Solutions Architect",
          "hsnSac": "998314",
          "quantity": 2,
          "unitPrice": 5000,
          "amount": 10000
        }
      ],
      "subtotal": 10000,
      "cgst": 900,
      "sgst": 900,
      "totalAmount": 11800,
      "currency": "INR",
      "status": "PROCESSED",
      "extractionConfidence": 97.8,
      "extractedFields": [],
      "anomalies": [],
      "createdAt": "2026-06-15T10:23:00Z",
      "updatedAt": "2026-06-15T10:25:30Z",
      "receivedOn": "2026-06-15T10:23:00Z",
      "s3RawKey": "invoices/INV-2026-A1B2C3D4/invoice.pdf",
      "source": "UPLOAD"
    }
  ],
  "total": 1,
  "hasMore": false
}
```

Note: When `status` is provided, the query uses the `status-index` GSI. Otherwise, a full table scan is performed (limited by `pageSize`).

GET /invoices/{invoiceId}

Retrieves complete details of a specific invoice along with its full audit history.

```
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/invoices/INV-2026-A1B2C3D4"
```

Path Parameters

Parameter	Required	Description
invoiceId	Yes	The ID of the target invoice (e.g., INV-2026-A1B2C3D4)

Success Response — 200 OK

```
{
  "invoice": {
    "invoiceId": "INV-2026-A1B2C3D4",
    "vendorId": "PENDING",
    "vendorName": "Acme Solutions Pvt. Ltd.",
    "vendorAddress": "123 MG Road, Bengaluru 560001",
    "gstIn": "29AABCU9603R1ZM",
    "invoiceNumber": "INV-2026-9812",
    "invoiceDate": "2026-06-15",
    "dueDate": "2026-07-15",
    "poNumber": "PO-4567",
    "invoiceType": "Standard",
    "lineItems": [
      {
        "description": "AWS Training - Solutions Architect",
        "hsnSac": "998314",
        "quantity": 2,
        "unitPrice": 5000,
        "amount": 10000
      }
    ],
    "subtotal": 10000,
    "cgst": 900,
    "sgst": 900,
    "totalAmount": 11800,
    "currency": "INR",
    "status": "PROCESSED",
    "extractionConfidence": 97.8,
    "extractedFields": [
      {
        "fieldName": "vendorName",
        "extractedValue": "Acme Solutions Pvt. Ltd.",
        "confidence": 98.5,
        "validationStatus": "MATCHED"
      },
      {
        "fieldName": "gstIn",
        "extractedValue": "29AABCU9603R1ZM",
        "confidence": 99.1,
        "validationStatus": "MATCHED"
      }
    ]
  }
}
```

```
],
"anomalies": [],
"approvedBy": "Admin User",
"approvalTimestamp": "2026-06-15T11:00:00Z",
"approvalComments": "Verified manually",
"assignedTo": "reviewer",
"createdBy": "web-user",
"createdAt": "2026-06-15T10:23:00Z",
"updatedAt": "2026-06-15T11:00:00Z",
"receivedOn": "2026-06-15T10:23:00Z",
"s3RawKey": "invoices/INV-2026-A1B2C3D4/invoice.pdf",
"s3AuditKey": "audit/INV-2026-A1B2C3D4/pipeline-report-1718445600000.json",
"s3ExtractedJsonKey": "textract/INV-2026-A1B2C3D4/raw-response.json",
"source": "UPLOAD"
},
"auditLogs": [
{
  "auditId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "invoiceId": "INV-2026-A1B2C3D4",
  "event": "Invoice uploaded via web dashboard",
  "eventType": "INGESTION",
  "timestamp": "2026-06-15T10:23:00Z",
  "user": "web-user",
  "details": "PDF \"invoice.pdf\" uploaded via React dashboard",
  "metadata": {
    "filename": "invoice.pdf",
    "size": "245760"
  }
},
{
  "auditId": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
  "invoiceId": "INV-2026-A1B2C3D4",
  "event": "Document extraction completed",
  "eventType": "EXTRACTION",
  "timestamp": "2026-06-15T10:23:15Z",
  "user": "system",
  "details": "Textract extracted 10 fields, 2 line items. Overall confidence:
97.8%",
  "metadata": {
    "fieldsCount": "10",
    "lineItemsCount": "2",
    "confidence": "97.8"
  }
},
{
  "auditId": "c3d4e5f6-a7b8-9012-cdef-123456789012",
  "invoiceId": "INV-2026-A1B2C3D4",
  "event": "AI validation completed",
  "eventType": "VALIDATION",
  "timestamp": "2026-06-15T10:23:30Z",
  "user": "system",
  "details": "Bedrock validation: PROCESSED. 0 anomalies detected. Confidence:
```

```

97.8%",
  "metadata": {
    "status": "PROCESSED",
    "anomaliesCount": "0",
    "anomalyTypes": "",
    "confidence": "97.8"
  }
}
]
}
}

```

Error Responses

Code	Body	Cause
400	{ "error": "Missing invoiceId" }	Empty or missing path parameter
404	{ "error": "Invoice not found" }	No record found in DynamoDB for the given invoiceId

Implementation note: The invoice is fetched using the composite key (`invoiceId`, `vendorId="PENDING"`). Audit logs are queried from the `AuditEntries` table using the `invoiceId`-index GSI.

⚠️ 5. Exceptions — GET /exceptions

Retrieves all invoices that have encountered anomalies or require manual review.

```

curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/exceptions"
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/exceptions?
pageSize=50"

```

Query Parameters

Parameter	Required	Default	Description
pageSize	No	25	Maximum results per status query

Success Response — 200 OK

```

{
  "items": [
    {
      "invoiceId": "INV-2026-B2C3D4E5",
      "vendorName": "Unknown Vendor",
      "totalAmount": 5400,
      "status": "EXCEPTION",
      "anomalies": [
        {
          "type": "VENDOR_NOT_FOUND",
          "description": "Vendor name is missing or could not be extracted from the
invoice.",
          "severity": "HIGH"

```

```

    },
    {
      "type": "MISSING_GSTIN",
      "description": "GSTIN field is empty or not detected in the invoice.",
      "severity": "HIGH"
    }
  ],
  "extractionConfidence": 72.3
}
],
"total": 70,
"stats": {
  "totalExceptions": 70,
  "pendingReview": 24,
  "inProgress": 18,
  "resolved": 28
}
}

```

Field	Type	Description
items	InvoiceRecord[]	Combined list of invoices from 4 status categories (see below)
total	number	Total count of all exception-related invoices
stats.totalExceptions	number	Same as total
stats.pendingReview	number	Count of PENDING_REVIEW invoices
stats.inProgress	number	Count of IN_REVIEW invoices
stats.resolved	number	Count of RESOLVED invoices

Implementation note: The endpoint queries 4 statuses in parallel — `EXCEPTION` , `IN_REVIEW` , `PENDING_REVIEW` , and `RESOLVED` — and concatenates the results. Each query is independently limited by `pageSize` .

6. Audit Logs — GET /audit

Queries the audit log table for a specific invoice's complete event history.

```
curl "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/audit?invoiceId=INV-2026-A1B2C3D4"
```

Query Parameters

Parameter	Required	Description
invoiceId	Yes	The target invoice ID

Success Response — `200 OK`

```
{
  "entries": [
    {
      "auditId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
      "invoiceId": "INV-2026-A1B2C3D4",
      "event": "Invoice uploaded via web dashboard",
      "eventType": "INGESTION",
      "timestamp": "2026-06-15T10:23:00Z",
      "user": "web-user",
      "details": "PDF \"invoice.pdf\" uploaded via React dashboard",
      "metadata": {
        "filename": "invoice.pdf",
        "size": "245760"
      }
    }
  ]
}
```

Audit Event Types

Event Type	Description
INGESTION	Invoice received (via email or upload)
EXTRACTION	Textract/Bedrock document extraction completed
VALIDATION	Bedrock AI validation completed
APPROVAL	Invoice approved by a reviewer
REJECTION	Invoice rejected by a reviewer
REPROCESS	Invoice sent back for reprocessing
PERSISTENCE	Final pipeline results persisted to DynamoDB + S3
ERROR	Pipeline error occurred at any stage

Error Responses

Code	Body	Cause
400	{ "error": "Missing invoiceId query parameter" }	No invoiceId provided

 7. Workflow Actions — POST /approvals

Submits a review decision from the React dashboard. Handles dynamic corrections to extracted fields.

Lambda: approval-handler-{env}

```
curl -X POST "https://<api-id>.execute-api.<region>.amazonaws.com/<env>/approvals" \
-H "Content-Type: application/json" \
-d '{
```

```

    "invoiceId": "INV-2026-A1B2C3D4",
    "action": "APPROVE",
    "taskToken": "AAAAKgAAAAIA...",
    "reviewer": "Admin User",
    "comments": "Fixed subtotal mismatch",
    "correctedFields": {
      "INV-2026-A1B2C3D4-Subtotal": "500.00",
      "INV-2026-A1B2C3D4-Vendor Name": "Acme Corp"
    }
  }
}'

```

Request Body

Field	Type	Required	Description
invoiceId	string	Yes	Target invoice ID
action	string	Yes	One of: APPROVE, REJECT, REPROCESS
taskToken	string	No	Step Functions task token. If omitted, the handler automatically retrieves it from DynamoDB (looked up by invoiceId with vendorId = "PENDING").
reviewer	string	No	Name of the reviewer performing the action
comments	string	No	Review comments
correctedFields	Record<string, string>	No	Field corrections (see format below)

correctedFields Key Format

Keys **must** be prefixed with the invoice ID followed by a hyphen:

```
{invoiceId}-{fieldName}
```

Supported field names and their DynamoDB mappings:

correctedFields Key Suffix	DynamoDB Field	Value Parsing
Vendor Name	vendorName	Stored as string
GSTIN	gstIn	Stored as string
Invoice Number	invoiceNumber	Stored as string
Invoice Date	invoiceDate	Stored as string
Due Date	dueDate	Stored as string
PO Number	poNumber	Stored as string
Total Amount	totalAmount	Parsed as float (non-numeric chars stripped)

Subtotal	subtotal	Parsed as float (non-numeric chars stripped)
tax	cgst	Stored as string

Important: Only keys that start with the correct `invoiceId-` prefix are processed. Others are silently ignored.

Action Behaviors

Action	DynamoDB Status	Step Functions	Anomalies	Audit Event
APPROVE	→ PROCESSED	SendTaskSuccess (non-blocking — token expiry is tolerated)	Cleared (1)	APPROVAL
REJECT	→ RESOLVED	SendTaskFailure with error REJECTED (non-blocking)	Cleared (1)	REJECTION
REPROCESS	→ IN_PROGRESS	— (no Step Functions interaction)	Cleared (1)	REPROCESS

Note on expired tokens: If the Step Functions task token has expired or is invalid, the handler logs a warning and proceeds with the DynamoDB update. The approval/rejection is **not blocked** by a stale token.

Success Response — 200 OK

```
{
  "success": true,
  "invoiceId": "INV-2026-A1B2C3D4",
  "action": "APPROVE",
  "message": "Invoice approved successfully"
}
```

Error Responses

Code	Body	Cause
400	{ "error": "Missing invoiceId or action" }	Missing required body fields
500	{ "error": "Approval action failed", "message": "..." }	DynamoDB update or audit write failure

Side Effects

1. DynamoDB record updated with new status, reviewer info, and any corrected field values
2. AuditEntry written to the audit table
3. SNS notification published to InvoicePipelineNotifications-{env} topic (non-blocking — failure is logged but does not cause a 500)

Data Models (TypeScript Schemas)

InvoiceRecord

The complete DynamoDB record for an invoice. Returned by `GET /invoices` , `GET /invoices/{invoiceId}` ,and `GET /exceptions` .

```
interface InvoiceRecord {
  // Keys
  invoiceId: string;          // PK – format: "INV-{YEAR}-{8-CHAR-UUID}"
  vendorId: string;           // SK – set to "PENDING" during processing

  // Core fields
  vendorName: string;
  vendorAddress?: string;
  gstin?: string;             // 15-character Indian GSTIN
  invoiceDate: string;
  dueDate?: string;
  poNumber?: string;
  invoiceNumber: string;
  invoiceType?: string;

  // Financial
  lineItems: LineItem[];
  subtotal: number;
  cgst: number;               // Central GST
  sgst: number;               // State GST
  totalAmount: number;
  currency: string;           // Default: "INR"

  // Processing metadata
  status: InvoiceStatus;
  extractionConfidence: number; // 0–100
  extractedFields: ExtractedField[];
  anomalies: Anomaly[];

  // Approval
  approvedBy?: string;
  approvalTimestamp?: string;
  approvalComments?: string;
  assignedTo?: string;

  // Audit & traceability
  createdBy?: string;
  createdAt: string;          // ISO 8601
  updatedAt: string;          // ISO 8601
  receivedOn: string;         // ISO 8601

  // S3 references
  s3RawKey: string;
  s3AuditKey?: string;
  s3ExtractedJsonKey?: string;

  // Step Functions
  taskToken?: string;         // Stored for dashboard-initiated approvals
```

```

    executionArn?: string;

    // Source
    source: 'EMAIL' | 'UPLOAD';
    sourceEmail?: string;
}

```

Supporting Types

```

type InvoiceStatus =
  | 'IN_PROGRESS'
  | 'PROCESSED'
  | 'IN_REVIEW'
  | 'PENDING_REVIEW'
  | 'EXCEPTION'
  | 'RESOLVED';

type ExceptionType =
  | 'MISSING_GSTIN'
  | 'AMOUNT_MISMATCH'
  | 'VENDOR_NOT_FOUND'
  | 'DUPLICATE_INVOICE'
  | 'LOW_CONFIDENCE_SCORE';

interface LineItem {
  description: string;
  hsnSac: string;           // HSN/SAC code (Indian tax classification)
  quantity: number;
  unitPrice: number;
  amount: number;
}

interface ExtractedField {
  fieldName: string;
  extractedValue: string;
  confidence: number;       // 0-100
  validationStatus: 'MATCHED' | 'MISMATCH' | 'NOT_FOUND' | 'PENDING';
}

interface Anomaly {
  type: ExceptionType;
  description: string;
  severity: 'HIGH' | 'MEDIUM' | 'LOW';
  suggestedValue?: string;
}

interface AuditEntry {
  auditId: string;           // PK - UUID
  invoiceId: string;        // GSI PK
  event: string;
  eventType: AuditEventType;
}

```

```
timestamp: string;           // ISO 8601 – also serves as GSI SK
user: string;
details: string;
metadata?: Record<string, string>;
}

type AuditEventType =
  | 'INGESTION'
  | 'EXTRACTION'
  | 'VALIDATION'
  | 'APPROVAL'
  | 'REJECTION'
  | 'REPROCESS'
  | 'PERSISTENCE'
  | 'ERROR';

// POST /approvals request body
interface ApprovalCallbackPayload {
  invoiceId: string;
  action: 'APPROVE' | 'REJECT' | 'REPROCESS';
  taskToken: string;
  reviewer: string;
  comments?: string;
  correctedFields?: Record<string, string>;
}
```

DynamoDB Table Reference

InvoiceRecords-{env}

Attribute	Type	Role
invoiceId	S	Partition Key (PK)
vendorId	S	Sort Key (SK) — set to "PENDING" during processing
status	S	GSI PK for status-index
updatedAt	S	GSI SK for status-index
invoiceNumber	S	GSI PK for invoiceNumber-index (used for duplicate detection)

Global Secondary Indexes:

Index Name	PK	SK	Projection	Used By
status-index	status	updatedAt	ALL	GET /invoices?status=, GET /dashboard, GET /exceptions
invoiceNumber-index	invoiceNumber	—	ALL	Duplicate invoice detection (Bedrock validator)

AuditEntries-{env}

Attribute	Type	Role
auditId	S	Partition Key (PK)
timestamp	S	Sort Key (SK)
invoiceId	S	GSI PK for invoiceId-index

Global Secondary Indexes:

Index Name	PK	SK	Projection	Used By
invoiceId-index	invoiceId	timestamp	ALL	GET /audit?invoiceId=, GET /invoices/{invoiceId} (audit logs)

 Internal Contracts (Step Functions)

The approval-handler Lambda can also be invoked **directly** by Step Functions (not via API Gateway) to send approval emails. This happens during the HumanApproval state of the pipeline.

Direct Invocation Payload

```
{
  "action": "SEND_APPROVAL_EMAIL",
  "invoiceId": "INV-2026-A1B2C3D4",
  "vendorName": "Acme Solutions Pvt. Ltd.",
  "totalAmount": 11800,
  "anomalies": ["VENDOR_NOT_FOUND", "MISSING_GSTIN"],
  "taskToken": "AAAAKgAAAAIA...",
  "assignedTo": "reviewer",
  "assignedToEmail": "reviewer@example.com"
}
```

Behavior

- 1. Stores the taskToken in DynamoDB (so the dashboard can later use it for POST /approvals without needing the token)
- 2. Discovers the API Gateway URL automatically (queries the API Gateway REST APIs, looking for InvoicePipelineAPI-{env})
- 3. Sends an HTML email via SES with Approve and Reject links pointing to GET /approve and GET /reject
- 4. Returns { "success": true, "message": "Approval email sent successfully" }

Graceful degradation: If SES email delivery fails (e.g., unverified sandbox address), the error is logged but the function does **not** throw. Approval can still be performed from the dashboard UI.